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References:

Chandrika Prasad, Advance Engineering Mathematics, Stanford Publishers and Distributors

Engineering Mathematics-T. Veerarajan, TMH

Faculty

HOD

DEAN ACADEMICS

Handwritten signature and date: 20/10/2022

UNIT-5 (ADVANCE ENGINEERING MATHEMATICS)

FINITE DIFFERENCES OPERATORS

LECTURE NO:-34

Finite Difference

1. Finite Difference

Great scientist and mathematician Newton introduced the calculus of Finite Differences. With the rapid development of numerical answer to practical problem, numerical method have become very important and useful in Engg., Technology, Applied mathematics etc.

The process of finding the result in numerical form by computing method of statistical data is called numerical analysis. The given data are called input information from input is called the algorithm and numerical result obtained is called output information.

input—algorithm output

In numerical analysis, calculus of finite differences and operators are of fundamental importance.

In calculus of finite differences, the corresponding values of the function $y=f(x)$ are studied which are known as entry and the independent variable x is called argument.

Operators:

1. Forward Difference Operator(Δ):

Let $y=f(x)$ be a function of equal spaces of the independent variable x . The order forward difference of y is change in y when x is increased by a positive difference h .

This operation is carried out through the notation Δ . Thus

$$\Delta f(x) = f(x+h) - f(x)$$

$$\text{Or } \Delta y_0 = y_1 - y_0$$

Here h is known as differencing. Δ is called forward difference operator.

In general $\Delta y_r = y_{r+1} - y_r$ ($r=0,1,2,3,\dots,n-1$)

Similarly $\Delta f(x+h) = f(x+2h) - f(x+h)$

The second order forward difference are defined by

$$\Delta^2 f(x) = \Delta(\Delta f(x))$$

$$= \Delta(f(x+h) - f(x))$$

$$= \Delta f(x+h) - \Delta f(x)$$

$$= f(x+2h) - f(x+h) - f(x+h) + f(x)$$

And $\Delta^2 y_r = y_{r+2} - 2y_{r+1} + y_r$, $r=0,1,2,3,\dots,n-2$

In general $\Delta^n y_0 = y_n - {}^n C_1 y_{n-1} + \dots + (-1)^n y_0$

Forward Difference Table

Argument(x)	Entry(y)	First Difference		Second Difference
X_0	y_0	Δy_0	$\Delta^2 y_0$	
X_0+h	y_1	Δy_1	$\Delta^2 y_1$	
X_0+2h	y_2	Δy_2		
X_0+3h	y_3			

Forward Finite Difference of a Polynomial:

Consider a polynomial of n^{th} degree in x as

$$F(x) = a_0 x^n + a_1 x^{n-1} + \dots + a_n$$

$$\text{Thus } \Delta^n f(x) = a n(n-1) \dots 2 \cdot 1 \cdot h^n = a \cdot n! h^n$$

$$\Delta^{n+1} f(x) = 0$$

2. Backward Difference Operator(∇):

The first order backward difference of y is the change in y when x is increased by a positive difference h . This operation is carried out by the symbol ∇ , Thus

$$\nabla f(x) = f(x) - f(x-h), (h > 0)$$

$$\nabla y_r = y_r - y_{r-1}, \nabla y_0 = y_1 - y_0, \nabla y_1 = y_2 - y_1,$$

$$\nabla^2 y_r = y_r - 2y_{r-1} + y_{r-2}$$

$$\nabla^2 f(x) = f(x) - 2f(x-h) + f(x-2h)$$

Backward Difference Table

x	y	∇y	$\nabla^2 y$	$\nabla^3 y$
X_0	y_0			
X_0+h	y_1	∇y_1		
X_0+2h	y_2	∇y_2	$\nabla^2 y_2$	
X_0+3h	y_3	∇y_3	$\nabla^2 y_3$	$\nabla^3 y_3$

3. Central Difference Operator(δ)

The forward and backward differences are mainly useful in interpolating the values near and beginning and the end of the table respectively. Central differences are particularly useful in interpolation for values of x in the interior of the table. It is denoted by δ and the first order central difference is defined as

$$\delta f(x) = f\left(x + \frac{h}{2}\right) - f\left(x - \frac{h}{2}\right)$$

$$\delta y_r = y_{r+1/2} - y_{r-1/2} \quad r=0, \pm 1, \pm 2, \dots, \pm n$$

$$\delta^2 y_r = \delta(\delta y_r) = \delta\left(y_{r+\frac{1}{2}} - y_{r-\frac{1}{2}}\right)$$

$$= \delta y_{r+\frac{1}{2}} - \delta y_{r-\frac{1}{2}}$$

$$= y_{r+1} - 2y_r + y_{r-1}$$

Central Difference Table

X	y	δy	$\delta^2 y$	$\delta^3 y$
X_0-2h	y_{-2}			
X_0-h	y_{-1}	$\delta y_{-3/2}$		
X_0	y_0	$\delta y_{-1/2}$	$\delta^2 y_{-1}$	
X_0+h	y_1	$\delta y_{1/2}$	$\delta^2 y_0$	$\delta^3 y_{-1/2}$
X_0+2h	y_2	$\delta y_{3/2}$	$\delta^2 y_1$	$\delta^3 y_{1/2}$

4. Shift Operator(E):.

When value of x are given at a regular interval h, then the shift operator E is defined as

$$E\{f(x)\} = f(x+h) \text{ or } Ey_r = y_{r+1}$$

The higher order operations of E can be defined as

$$E^2 y_r = y_{r+2}, E^m y_r = y_{r+m}$$

$$\text{Also } E^{-1} y_{r+1} = y_r$$

Thus the inverse operator E^{-1} when operated on y gives the value of the function when x is

increased by h.

$$\nabla y_r = y_r - y_{r-1}$$

$$\nabla y_r = y_r - E^{-1} y_r$$

$$\nabla = 1 - E^{-1}$$

$$\Delta y_r = y_{r+1} - y_r$$

$$\Delta y_r = E y_r - y_r$$

$$\Delta = E - 1$$

Therefore we have $\Delta = E\nabla = \nabla E$

Similarly $\delta = E^{1/2} - E^{-1/2}$

$$\mu = 1/2(E^{\frac{1}{2}} + E^{-\frac{1}{2}})$$

5. Average Operator(μ):

It is defined as

$$\mu f(x) = 1/2 \left[f\left(x + \frac{1}{2}\right) + f\left(x - \frac{1}{2}\right) \right]$$

$$\mu y_r = 1/2 (y_{r+\frac{1}{2}} + y_{r-\frac{1}{2}})$$

$$\mu y_r = 1/2 (E^{1/2} y_r + E^{-1/2} y_r)$$

Or

$$\mu = 1/2 (E^{1/2} + E^{-1/2})$$

References:

Advance Engineering Mathematics - H.K.Das, S. Chand

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UNIT-5 (ADVANCE ENGINEERING MATHEMATICS)

RELATIONSHIP BETWEEN OPERATORS

LECTURE NO:-35

Relationship Between Operators

Lot of useful and interesting results can be derived among the operators discussed above. First of all, we determine the relation between forward and backward difference operators.

$$\Delta y_i = y_{i+1} - y_i$$

$$\nabla y_{i+1} = \delta y_{i+1/2}$$

$$\Delta^2 y_i = y_{i+2} - 2y_{i+1} + y_i$$

$$\nabla^2 y_{i+2} = \delta^2 y_{i+1} \text{ etc.}$$

In general, $\Delta^n y_i$

$$\nabla^n y_{i+n}, i = 0, 1, 2, \dots$$

There is a good relation between E and Δ operators.

$$\Delta f(x) = f(x+h) - f(x) = Ef(x) - f(x) = (E-1)f(x).$$

From this relation one can conclude that the operators Δ and $E-1$ are equivalent. That is, $\Delta \equiv E-1$ or $E \equiv \Delta+1$. The relation between ∇ and E operators is derived below: $\nabla f(x) = f(x) - f(x-h) = f(x) - E^{-1}f(x) = (1-E^{-1})f(x)$. That is, $\nabla \equiv 1-E^{-1}$. The expression for higher order forward differences i

$$1. \Delta \equiv E-1$$

Proof: From the definition of Δ we know that

$$\Delta f(x) = f(x+h) - f(x) \text{ and}$$

$$E[f(x)] = f(x+h)$$

where h is the interval of difference.

$$\Delta f(x) = f(x+h) - f(x)$$

$$\Delta f(x) = Ef(x) - f(x)$$

$$\Rightarrow \Delta f(x) = (E-1)f(x)$$

$$\Delta \equiv E-1$$

$$\therefore E \equiv 1 + \Delta$$

$$2. \Delta E \equiv \Delta E$$

Proof:

$$E(\Delta f(x)) = E[f(x+h) - f(x)]$$

$$= E f(x+h) - E f(x)$$

$$= f(x+2h) - f(x+h)$$

$$= \Delta f(x+h)$$

$$= \Delta E f(x)$$

$$\Delta E \equiv \Delta E$$

$$3. \nabla \equiv E^{-1} - 1 / E$$

$$\nabla f(x) = f(x) - f(x-h)$$

$$= f(x) - E^{-1} f(x)$$

$$= (1 - E^{-1}) f(x)$$

$$\nabla \equiv (1 - E^{-1})$$

$$\nabla \equiv 1 - 1/E$$

$$\text{Hence } \nabla \equiv [E - 1]/E$$

$$4. \delta = E^{1/2} - E^{-1/2}$$

$$\delta f(x) = f(x+h/2) - f(x-h/2)$$

$$= E^{1/2} f(x) - E^{-1/2} f(x)$$

$$= (E^{1/2} - E^{-1/2}) f(x)$$

$$\delta = E^{1/2} - E^{-1/2}$$

$$(5) E = e^{hD}$$

Proof:- We know that by Taylor series

$$f(x+h) = f(x) + hf'(x) + \frac{h^2}{2!} f''(x) + \frac{h^3}{3!} f'''(x) + \dots$$

$$\text{or } Ef(x) = f(x) + hDf(x) + \frac{h^2}{2!} D^2 f(x) + \frac{h^3}{3!} D^3 f(x) + \dots$$

$$= \left[1 + \frac{hD}{1} + \frac{(hD)^2}{2!} + \frac{(hD)^3}{3!} + \dots \right] f(x)$$

$$= e^{hD} f(x)$$

$$\therefore E = e^{hD}$$

$$(6) D = \frac{1}{h} \sin^{-1}(\mu \delta)$$

Proof: we know that

$$E = e^{hD}$$

$$\text{and } D = \frac{1}{h} \left[\Delta - \frac{\Delta^2}{2} + \frac{\Delta^3}{3} - \frac{\Delta^4}{4} + \dots \right]$$

$$\text{also } \nabla = 1 - E^{-1}$$

$$\text{or } E^{-1} = 1 - \nabla$$

$$\text{i.e. } e^{-hD} = 1 - \nabla$$

Taking logarithm both the sides, we get.

$$-hD = \log(1 - \nabla)$$

$$d = \frac{1}{h} \left(\nabla + \frac{\nabla^2}{2} + \frac{\nabla^3}{3} + \dots \right)$$

$\therefore$ we know that

$$\begin{aligned} \sin(hd) &= \frac{e^{hd} - e^{-hd}}{2} \\ &= \frac{E - E^{-1}}{2} \\ &= \frac{1}{2} (E^{v_2} + E^{-v_2}) (E^{v_2} - E^{-v_2}) \end{aligned}$$

$$\sin hd = \mu\delta$$

$$\therefore D = \frac{1}{h} \sin^{-1} \mu\delta$$

$$(6) \quad \Delta \nabla = \nabla \Delta = \Delta - \nabla = \delta^2$$

proof :

$$\begin{aligned} \Delta \nabla f(x) &= \Delta [\nabla f(x)] \\ &= \Delta [f(x) - f(x-h)] \\ &= \Delta f(x) - \Delta f(x-h) \\ &= f(x+n) - f(x) - [f(x) - f(x-h)] \\ &= \Delta f(x) - \Delta f(x-h) \\ &= f(x+n) - f(x) - [f(x) - f(x-h)] \\ &= \nabla f(x) - \nabla f(x) \end{aligned}$$

$$\therefore \nabla \Delta = \nabla - \Delta$$

$$\begin{aligned} \text{and } \delta^2 f(x) &= [E^{v_2} - E^{-1/2}]^2 f(x) \\ &= [E + E^{-1} - 2] f(x) \\ &= Ef(x) + E^{-1}f(x) - 2f(x) \\ &= f(x+h) - f(x-h) - 2f(x) \end{aligned}$$

$$\delta^2 f(x) = [f(x+h) - f(x)] - [f(x) - f(x-h)]$$

$$\delta^2 = \Delta - \nabla$$

$$\text{therefore } \Delta \nabla = \nabla \Delta = \Delta - \nabla = \delta^2$$

$$(7) \quad \Delta + \nabla = \frac{\Delta}{\nabla} - \frac{\nabla}{\Delta}$$

$$\begin{aligned} \text{Proof } \frac{\Delta}{\nabla} - \frac{\nabla}{\Delta} &= \frac{\Delta^2 - \nabla^2}{\Delta \nabla} \\ &= \frac{(\Delta - \nabla)(\Delta + \nabla)}{\Delta - \nabla} \\ &= \frac{(\Delta - \nabla)(\Delta + \nabla)}{\Delta - \nabla} (\because \Delta \nabla = \Delta - \nabla) \\ &= \Delta + \nabla \end{aligned}$$

$$(8) \quad 1(1 + \Delta)(1 - \nabla) = 1$$

$$\begin{aligned} \text{Proof } (1 + \Delta)(1 - \nabla) &= 1 + \Delta - \nabla - \Delta \nabla \\ &= 1 + \Delta \nabla - \Delta \nabla \\ &= 1 \end{aligned}$$

$$(9) \quad \nabla E = \Delta = E \nabla$$

Proof take now

$$\begin{aligned} \nabla E f(x) &= \nabla [E f(x)] \\ &= [1 - E^{-1}] E f(x) \\ &= \Delta f(x) \\ \therefore \nabla E &= \Delta \end{aligned}$$

again $E\nabla f(x) = E[\nabla f(x)]$
 $\therefore E\nabla = E(1 - E^{-1})f(x)$
 $= \Delta f(x)$

(10) $\nabla^r y_{k+r} = \Delta^r y_k$

Proof $\nabla^r y_{k+r} = (\Delta E^{-1})^r y_{k+r}$
 $= \Delta^r E^{-r} y_{k+r}$
 $= \Delta^r E^{-r} [y_{k+r}]$
 $= \Delta^r y_k$

References:

Advance Engineering Mathematics - H.K. Das, S. Chand

UNIT-5 (ADVANCE ENGINEERING MATHEMATICS)

INTERPOLATION BY NEWTON'S FORWARD BACKWARD DIFFERENCE FORMULA

LECTURE NO:-36

Interpolation by Newton'S Forward Backward Difference Formula

Suppose we are given the following values of $y = f(x)$ for a set of values of x : $x: x_0, x_1, x_2, \dots, x_n$; $y: y_0, y_1, y_2, \dots, y_n$. Then the process of finding the value of y corresponding to any value of $x = x_i$ between x_0 and x_n is called interpolation. Thus interpolation is the technique of estimating the value of a function for any intermediate value of the independent variable while the process of computing the value of the function outside the given range is called extrapolation. The term interpolation however, is taken to include extrapolation. If the function $f(x)$ is known explicitly, then the value of y corresponding to any value of x can easily be found. Conversely, if the form of $f(x)$ is not known (as is the case in most of the applications), it is very difficult to determine the exact form of $f(x)$ with the help of tabulated set of values (x_i, y_i) . In such cases, $f(x)$ is replaced by a simpler function $\phi(x)$ which assumes the same values as those of $f(x)$ at the tabulated set of points. Any other value may be calculated from $\phi(x)$ which is known as the interpolating function or smoothing function. If $\phi(x)$ is a polynomial, then it is called the interpolating polynomial and the process is called the polynomial interpolation. Similarly when $\phi(x)$ is a finite trigonometric series, we have trigonometric interpolation. But we shall confine ourselves to polynomial interpolation only. The study of interpolation is based on the calculus of finite differences. We begin by deriving two important interpolation formulae by means of forward and backward differences of a function. These formulae are often employed in engineering and scientific investigations.

Newton Forward And Backward Interpolation

Interpolation is the technique of estimating the value of a function for any intermediate value of the independent variable, while the process of computing the value of the function outside the given range is called **extrapolation**.

Forward Differences: The differences $y_1 - y_0, y_2 - y_1, y_3 - y_2, \dots, y_n - y_{n-1}$ when denoted by $\Delta y_0, \Delta y_1, \Delta y_2, \dots, \Delta y_{n-1}$ are respectively, called the first forward differences. Thus, the first forward differences are :

NEWTON'S GREGORY BACKWARD INTERPOLATION FORMULA :

This formula is useful when the value of $f(x)$ is required near the end of the table. h is called the interval of difference and $u = (x - x_n) / h$, Here x_n is last term.

Interpolation for equi-spaced valued:

In performing experiment we quite often come across function of a variable y which tabulated at regular interval of x . If we need the value of such a function for a values of x other than those tabulated. It is done by interpolation. for interpolation, we may take the differences of two successive value of the function, and use proportional parts to obtain the intermediate values. To find the value of the function corresponding to the value of x laying outside the range of data is called extrapolation.

Newton's forward difference interpolation formula.

Let the function $y = f(x)$ take the values $y_0, y_1, y_2, \dots, y_n$ corresponding to the equi-distant values of x as $x_0, x_1 (= x_0 + h), x_2 (= x_0 + 2h), \dots, x_n (= x_0 + nh)$ thus, we have a set of $(n+1)$ pairs of values $(x_0, y_0), (x_1, y_1), \dots, (x_n, y_n)$

we can write $y_n = E^n y_0 = (1 + \Delta)^n y_0$

$$\text{or } y_n = y_0 + n\Delta y_0 + \frac{n(n-1)}{2}\Delta^2 y_0 + \dots + \Delta^n y_0$$

This derivation is true when n is positive integer our next assertion is that this formula is true for other than positive integral values of n also. Hence it is used for interpolatin between the tabular values i.e. for finding $y_n = f(x + nh)$ where n is fractional. while using this formula for interpolation, n should be taken between 0 and 1 (or between -1 and 0) as far as possible. For such values of n the coefficients, $n, \frac{n(n-1)}{2}, \frac{n(n-2)}{3}, \dots$ are small and the terms

decrease rapidly . then summing upto four or five terms may give sufficiently accurate. value of y_n

(i) The area of a circle of diameter d is given for the following value.

d :	80	85	90	95	100
A:	5026	5674	6363	7088	7854

find approximate value of the areas of circle of diameter 82.

Sol. forward difference table.

D	A-f(d)	$\Delta f(d)$	$\Delta^2 f(d)$	$\Delta^3 f(d)$	$\Delta^4 f(d)$
80	5026				
85	5674	648			
90	6362	688	40	-2	
95	7088	726	38	2	4
100	854	766	40		

here f (82) is to be determined , therefore $a+mh=82$

here $h=5$, $a=80$, therefore $80+m \cdot 5=82 \Rightarrow m=\frac{2}{5}$

Now by newton's forward interpolatin formula.

$$\begin{aligned}
 f(82) &= f(80) + \frac{2}{5} \left(\frac{2}{5} - 1 \right) \frac{1}{2} \Delta^2 f(80) \\
 &\quad + \frac{2}{5} \left(\frac{2}{5} - 1 \right) \left(\frac{2}{5} - 2 \right) \frac{1}{6} \Delta^3 f(80) + \dots \\
 &= 5026 + \frac{2}{5} \times 648 + \frac{2}{5} \left(-\frac{9}{5} \right) \times \left(\frac{40}{3} \right) + \frac{2}{5} \left(\frac{3}{5} \right) \left(-\frac{6}{1} \right) \left(\frac{2}{6} \right) \\
 &= 5026 + 259.2 - 4.8 - 0.128 \\
 &= 5280.105 - 5080 (\text{appear})
 \end{aligned}$$

(2) Find f (1.5) from the following table.

x :	1	2	3	4	5	6	7	8
y :	1	8	27	64	126	216	347	512

(3) find the value of $\sin 52^\circ$ from the given table.

Q°	45°	50°	55°	60°
$\sin \theta^\circ$	0.7071	0.7660	0.8192	0.8660

Sol: here $x = 52^\circ$, $h = 5^\circ$ and $x_0 = 45$

$$\therefore P = \frac{x - x_0}{h} = \frac{52 - 45}{5} = \frac{7}{5} = 1.4$$

forward difference table.

x^0	$10^4 y$	$10^4 \Delta y$	$10^4 \Delta^2 y$	$10^4 \Delta^3 y$
45	7071			
		589		
50	7660		-57	
		532		-7
55	8192		-64	
		468		
60	8660			

Now by for diffrence formula.

$$\begin{aligned}
 f(x_0 + ph) &= y_0 + P\Delta y_0 + \frac{P(P-1)}{2} \Delta^2 y_0 + \frac{P(P-1)(P-2)}{3} \Delta^3 y_0 + \dots \\
 &= 10^4 f(x) = 10^4 y_0 + 10^4 P\Delta y_0 + 10^4 \Delta^2 y_0 \frac{P(P-1)}{2} + 10^4 \frac{P(P-1)(P-2)}{3} \Delta^3 y_0 + \dots \\
 &= 10^4 f(x) = 7071 + (1.4)(589) + \frac{(1.4)(1.4-1)}{2} (-57) \\
 &\quad + \frac{(1.4)(1.4-1)(1.4-2)}{3!} (-7)
 \end{aligned}$$

NEWTON'S GREGORY BACKWARD INTERPOLATION FORMULA :

This formula is useful when the value of $f(x)$ is required near the end of the table. h is called the interval of difference and $u = (x - a_n) / h$, Here a_n is last term.

Newton backward interpolation is essentially the same as Newton forward interpolation except that backward differences are used • Backward differences are defined as: Zeroth order

backward difference First order backward difference Second order backward difference kth order backward difference

Newton formulae can be obtained by manipulating Taylor series

- Newton interpolating function is related to easily computed forward/backward differences
- Error is readily established and estimated from the difference table (as long as you have one more data point than used in interpolation)
- Newton interpolation through data points gives the same degree polynomial as Lagrange interpolation
- Newton interpolation is more efficient than others.

Newton's backward differences interpolation formula:

$$y(x) = a_0 + a_1(x - x_n) + a_2(x - x_n)(x - x_{n-1}) + a_3(x - x_n)(x - x_{n-1})(x - x_{n-2}) + \dots + a_n(x - x_n)(x - x_{n-1}) \dots (x - x_1)$$

and then assume that y and $y(x)$ agree at the tabulated points.

$$x_n, x_{n-1}, x_{n-2}, \dots, x_1, x_0, \dots, x_1, x_0 \text{ then}$$

$$a_0 = y_n, a_1 = \frac{y_n - y_{n-1}}{x_n - x_{n-1}} = \frac{\nabla y_n}{h}, a_2 = \frac{\nabla^2 y_n}{h^1 | 2}$$

$$a_n = \nabla^n y_n$$

Hence for $x = x_n + rh$, we have.

$$y(x) = y_n + r \nabla y_n + \frac{r(r+1)}{| 2} \nabla^2 y_n + \dots + \frac{r(r+1)(r+2) \dots (r+n-1)}{| n} \nabla^n y_n$$

This is newton -gregory backward difference interpolation formula and is useful for interpolation for value near the end of tabular values.

(i) Given the following data, evaluate $y(0.62)$

x	:	0.0	0.2	0.4	0.6	0.8
y	:	0.3989	0.3910	0.3683	0.3332	0.2897

Sol. difference table.

x	y	Δy	$\Delta^2 y$	$\Delta^3 y$	$\Delta^4 y$
0.0	0.3989	-0.0079		-0.0148	
0.2	0.3910	-0.0227			0.024
0.4	0.3683			-0.124	0.0016
0.6	0.3332	-0.351		-0.0084	
0.8	0.2897				

$$y(x_0 + nh + 4n) = y_n + \frac{4}{| 1} \nabla y_n + \frac{4(4+1)}{| 2} \nabla^2 y_n + \dots$$

$$a + nh = .8 \quad a + nh + hu = 62$$

$$n_4 = -.18, 4 = -0.9$$

(3) Find $f(1.28)$ by using Newton's Gregory backward interpolation formula for the table.

x	:	1.15	1.20	1.25	1.30
y	:	1.0723	1.0954	0.1180	1.1401

Sol here $x=1.28$, $h = 0.05$, $x_n = 1.30$

$$\because x = x_n + ph$$

$$\therefore p = \frac{x - x_n}{h}$$

$$= \frac{1.28 - 1.30}{0.05}$$

$$= -0.4$$

Backward difference table is.

x	y	∇y	$\nabla^2 y$	$\nabla^3 y$
1.15	1.0723			
		0.0231		
1.20	1.0954		-0.005	
		0.0226		0
1.25	1.1180		-0.005	
		0.0221		
1.30	1.401			

$$f(x_n + ph) = y_n + p\nabla y_n + \frac{p(p+1)}{2!} \nabla^2 y_n + \frac{p(p+1)(p+2)}{3!} \nabla^3 y_n + \dots$$

$$= 1.401 - 0.4(0.0221) + \frac{(-0.4)(-0.4+1)}{2!} (0.005)$$

$$= 1.401 - 0.00884 + 0.00006$$

$$= 1.13132$$

4. Find $f(0.15)$ using Newton backward difference table from the data

x	f(x)	∇f	$\nabla^2 f$	$\nabla^3 f$	$\nabla^4 f$
0.1	0.09983				
		0.09884			
0.2	0.19867		-0.00199		

		0.09685	-0.00156	
0.3	0.29552		-0.00355	0.00121
		0.0939		-0.00035
0.4	0.38942		-0.0039	
		0.09		
0.5	0.47943			

$$s = (x - x_n) / h = (0.15 - 0.5) / 0.1 = -3.5$$

$$f(0.15) = f_n + s\nabla f_n + \frac{s(s+1)}{2!} \nabla^2 f_n + \frac{s(s+1)(s+2)}{3!} \nabla^3 f_n + \frac{s(s+1)(s+2)(s+3)}{4!} \nabla^4 f_n$$

$$= .97943 + (-3.5) \cdot .09 + \frac{(-3.5)(-3.5+1)}{2!} (-.0039) + \frac{(-3.5)(-3.5+1)(-3.5+2)}{3!} (-.00035) + \frac{(-3.5)(-3.5+1)(-3.5+2)(-3.5+3)}{4!} (.00121)$$

$$= 0.97943 - 0.315 - 0.01706 + 0.000765625 + 0.00033086$$

$$= 0.14847$$

5. Population in 1925

Year (x):	1891	1901	1911	1921	1931
Population (y): (in thousands)	46	66	81	93	101

Solution:

x	y	∇y	$\nabla^2 y$	$\nabla^3 y$	$\nabla^4 y$
1891	46	20			
1901	66	15	-5		
1911	81	12	-3	2	
1921	93	8	-4	-1	-3
1931	101				

Value in 1925 is 96.8368

References:

Advance Engineering Mathematics - H.K. Das, S. Chand

UNIT-5 (ADVANCE ENGINEERING MATHEMATICS)

GAUSS'S FORWARD AND BACKWARD & STIRLING'S FORMULA

LECTURE NO: -37

Gauss's forward formula

∴ we know that newton's Gregory forward difference formula is

$$y_p = f(x_0 + ph) = f(x_0) + p\Delta f(x_0) + \frac{p(p-1)}{|2|} \Delta^2 f(x_0) + \frac{p(p-1)(p-2)}{|3|} \Delta^3 f(x_0) + \frac{p(p-1)(p-2)(p-3)}{|4|} \Delta^4 f(x_0) + \dots (1)$$

put $x_0 = 0$ and $h = 1$, in (1) We get

$$y_p = f(0) + p\Delta f(0) + \frac{p(p-1)}{|2|} \Delta^2 f(0) + \frac{p(p-1)(p-2)}{|3|} \Delta^3 f(0) + \frac{p(p-1)(p-2)(p-3)}{|4|} \Delta^4 f(0) + \dots (2)$$

we know $\Delta^3 f(-1) = \Delta^2 f(0) - \Delta^2 f(-1)$

$$= \Delta^2 f(0) = \Delta^3 f(-1) + \Delta^2 f(-1)$$

and $\Delta^4 f(-1) = \Delta^3 f(0) - \Delta^3 f(-1)$

$$= \Delta^3 f(0) = \Delta^4 f(-1) + \Delta^3 f(-1)$$

also $\Delta^5 f(-1) = \Delta^4 f(0) - \Delta^4 f(-1)$

$$= \Delta^4 f(0) = \Delta^5 f(-1) + \Delta^4 f(-1) \text{ etc.}$$

using these result in (2), we get

$$\begin{aligned} y_p &= f(0) + p\Delta f(0) + \frac{p(p-1)}{|2|} [\Delta^3 f(-1) + \Delta^2 f(-1)] + \frac{p(p-1)(p-2)}{|3|} \\ &\quad + \left\{ \frac{p(p-1)(p-2)}{|3|} + \frac{p(p-1)(p-2)(p-3)}{|4|} \right\} \Delta^4 f(-1) + \dots \\ &= y_p = f(p) = f(0) + p\Delta f(0) + p \frac{(p-1)}{|2|} \Delta^2 f(-1) + \frac{p(p+1)(p-1)}{|3|} \Delta^3 f(-1) \end{aligned}$$

$$+ \frac{p(p+1)(p-1)}{|3|} \Delta^3 f(-1) + \frac{p(p+1)(p-1)(p-2)}{|4|} \Delta^4 f(-1) + \dots$$

which is called gauss's forward interpolation formula in central notation the formula can be

$$\text{written as } y_p = y_0 + p\delta y_{1/2} + \frac{p(p-1)}{|2|} \delta^2 y_0 + \frac{p(p+1)(p-1)}{|3|} \delta^3 y_{1/2} \\ + \frac{p(p+1)(p-1)(p-2)}{|4|} \delta^4 y_0 + \dots$$

this formula is application when p lies between 0 and 1/2

Ex. Apply a central difference formula to obtain $f(32)$ given that

$$f(25)=0.2707, f(35)=0.3386, f(30)=0.3027 \text{ and}$$

$$f(40)=0.3794$$

Sol. here $x_0 + ph = 32, h = 5$

Now taking arigin at 30

$$x_0 = 30 \text{ and } p = \frac{32 - x_0}{h} = \frac{32 - 30}{5} = 0.4$$

Difference Table.

P	x	$f(x)$	$\Delta f(x)$	$\Delta^2 f(x)$	$\Delta^3 f(x)$
-1	25	0.2707	0.032		
0	30	0.3027		0.0039	
			0.0359		0.0010
1	35	0.3386		0.0049	
			0.0408		
2	40	0.3794			

$$f(p) = f(0) + p\Delta f(0) + \frac{p(p-1)}{|2|} \Delta^2 f(-1) + \frac{p(p-1)(p+1)}{|3|} \Delta^3 f(-1)$$

$$f(0.4) = 0.3027 + (0.4)(0.0359) + \frac{(0.4)(0.4-1)}{2} (0.0039)$$

$$+ \frac{(0.4)(0.4+1)(0.4-1)}{6} (0.0010)$$

$$= 0.316536$$

Gauss's Backward Method:

The gaussian interpolation comes under the Central Difference Interpolation Formulae. Suppose we are given the following value of $y=f(x)$ for a set values of x :

$X: x_0, x_1, x_2, \dots, x_n$

$Y: y_0, y_1, y_2, \dots, y_n$

The differences $y_1 - y_0, y_2 - y_1, y_3 - y_2, \dots, y_n - y_{n-1}$ when denoted by $\Delta y_0, \Delta y_1, \Delta y_2, \dots, \Delta y_{n-1}$ are respectively, called the first forward differences. Thus the first forward differences are :

$$\Delta y_0 = y_1 - y_0$$

	.					
	.					
	.					
	x_{-2}	y_{-2}				
			Δy_{-1}			
	x_{-1}	y_{-1}		$\Delta^2 y_0$		
			Δy_0		$\Delta^3 y_1$	
	x_0	y_0		$\Delta^2 y_1$		$\Delta^4 y_0$
			Δy_1		$\Delta^3 y_2$	
	x_1	y_1		$\Delta^2 y_2$		
			Δy_2			
	x_2	y_2				
	.					
	.					
	.					

And after the creating table we calculate the value on the basis of following Gauss's Backward interpolation formula:

$$y_p = y_0 + p\Delta y_{-1} + \frac{p(p+1)}{2!} \Delta^2 y_{-1} + \frac{((p+1)p(p-1))}{3!} \Delta^3 y_{-2} + \dots$$

Problem:

From the following table, find the value of $e^{1.17}$ using Gauss's Backward formula.

x	1.00	1.05	1.10	1.15	1.20	1.25	1.30
e^x	2.7183	2.8577	3.0042	3.1582	3.3201	3.4903	3.6693

Solution:

We have

$$y_p = y_0 + p\Delta y_{-1} + \frac{p(p+1)}{2!}\Delta y_{-2} + \frac{((p+1)p(p-1))}{3!}\Delta y_{-3} + \dots$$

$$\text{where } p = (x_{1.17} - x_{1.15}) / h$$

$$\text{and } h = x_1 - x_0 = 0.05$$

$$\text{so, } p = 0.04$$

Now, we need to calculate $\Delta y_0, \Delta y^2_0, \Delta y^3_0 \dots$ etc.

	Index	X	Y	ΔY	$\Delta^2 Y$	$\Delta^3 Y$	$\Delta^4 Y$	$\Delta^5 Y$	$\Delta^6 Y$
	-3	1	2.718						
				0.139					
	-2	1.05	2.858		0.007				
				0.147		0.0004			
	-1	1.1	3.004		0.007		0		
				0.154		0.0004		0	
	0	1.15	3.158		0.008		0		0.0001
				0.162		0.0004		0.0001	
	1	1.2	3.32		0.008		0.0001		
				0.17		0.0005			
	2	1.25	3.49		0.009				
				0.179					
	3	1.3	3.669						

Put the required values in the formula-

$$y_{x=1.17} = 3.158 + (2/5)(0.154) + (2/5)(2/5 + 1)/2 \cdot (0.008) \dots$$

$$y_{x=1.17} = 3.2246$$

STIRLING FORMULA

Stirling's formula, also called **Stirling's approximation**, in analysis, a method for approximating the value of large factorials (written $n!$; e.g., $4! = 1 \times 2 \times 3 \times 4 = 24$) that uses the mathematical constants e (the base of the natural logarithm) and π . The formula is given by $n! \cong \left(\frac{n}{e}\right)^n \sqrt{2\pi n}$.

The Scottish mathematician James Stirling published his formula in *Methodus Differentialis sive Tractatus de Summatione et Interpolatione Serierum Infinitarum* (1730; "Differential Method with a Tract on Summation and Interpolation of Infinite Series"), a treatise on infinite series, summation, interpolation, and quadrature.

Stirling's formula is obtained by average of Gauss's forward and Gauss's backward interpolation formulas. So Stirling formula is a central difference formula to solve problems and using for the range $-1/4$ to $1/4$. This is best in all central difference formulas. So used often with conditions.

Stirling's interpolation formula looks like:

$$P_{2n}(x) = y_0 + q \cdot \frac{\Delta y_{-1} + \Delta y_0}{2} + \frac{q^2}{2!} \cdot \Delta^2 y_{-1} + \frac{q(q^2-1)}{3!} \cdot \frac{\Delta^3 y_{-2} + \Delta^3 y_{-1}}{2} +$$

$$+ \frac{q^2(q^2-1)}{4!} \cdot \Delta^4 y_{-2} + \frac{q(q^2-1)(q^2-2^2)}{5!} \cdot \frac{\Delta^5 y_{-3} + \Delta^5 y_{-2}}{2} +$$

$$+ \frac{q^2(q^2-1)(q^2-2^2)}{6!} \cdot \Delta^6 y_{-3} + \dots + \frac{q(q^2-1)(q^2-2^2)(q^2-3^2) \dots [q^2-(n-1)^2]}{(2n-1)!} \times$$

$$\times \frac{\Delta^{2n-1} y_{-n} + \Delta^{2n-1} y_{-(n-1)}}{2} + \frac{q^2(q^2-1)(q^2-2^2) \dots [q^2-(n-1)^2]}{(2n)!} \Delta^{2n} y_{-n},$$

$$q = \frac{x - x_0}{h}$$

where, as before,

There are also Gauss's, Bessel's, Lagrange's and others interpolation formulas. Formula is deduced with use of Gauss's first and second interpolation formulas.

Example :

Using Stirling's formula compute $f(12.2)$ from the data

x	f(x)	X
10	0.23967	-2
11	0.28060	-1
12	0.31788	0
13	0.35209	1
14	0.38368	2

X	$f_x = 10^5 f(x)$	δf_x	$\delta^2 f_x$	$\delta^3 f_x$	$\delta^4 f_x$
-2	23967				
		4093			
-1	28060		-365		
		3728		58	
0	31788		-307		-13
		3421		45	
1	35209		-262		

$$\begin{array}{r} 2 \qquad \qquad 3159 \\ 38368 \end{array}$$

$$f_{0.2} = f_0 + \binom{x}{1} \mu \delta f_0 + \frac{x}{2} \binom{x}{1} \delta^2 f_0 + \binom{x+1}{3} \mu \delta^3 f_0 + \frac{x}{4} \binom{x+1}{3} \delta^4 f_0$$

$$\begin{aligned} &= 31788 + 0.2 \frac{3728 + 3421}{2} + \frac{0.2 \times 0.2}{2} (-307) + \frac{(1.2)(0.2)(-0.8)}{3!} \frac{58 + 45}{2} + \frac{0.2}{4} \frac{1.2 \times 0.2 \times (-0.8)}{3!} (-13) \\ &= 31788 + 714.9 - 6.14 - 1.648 + 0.208 \\ &= 32495 \end{aligned}$$

$$\Rightarrow f(x) = 10^{-5} f_x = 0.32495$$

Advantages :

1. Stirling's formula decrease much more rapidly than other difference formulae hence considering first few number of terms itself will give better accuracy.
2. Forward or backward difference formulae use the onside information of the function where as Stirling's formula uses the function values on both sides of $f(x)$.

Example :

Compute $344.5^{1/3}$ for the equation $f(x) = x^{1/3}$

x	$u_x = 10^5 f(x)$	δu_x	$\delta^2 u_x$
342	6993191		
		6809	
343	7000000		-13
		6796	
344	7006796		-13
		6783	
345	7013579		-13
		6770	
346	7020349		-13
		6757	
347	7027106		

$$u_{1/2} = \frac{14020375}{2} - \frac{1}{8}(-13) = 7010189$$

$$\Rightarrow f(x) = 7.010189$$

References:

Advance Engineering Mathematics - H.K. Das, S. Chand

UNIT-5 (ADVANCED ENGINEERING MATHEMATICS)

LAGRANGE'S INTERPOLATION FOR UNEQUAL INTERVALS

LECTURE NO:-38

Lagrange's Interpolation For Unequal Intervals

Interpolation with Unequal Interval:

$$F(x) = \frac{(x-x_1)(x-x_2)\dots(x-x_n)}{(x_0-x_1)(x_0-x_2)\dots(x_0-x_n)} f(x_1) + \dots + \frac{(x-x_0)(x-x_1)\dots(x-x_{n-1})}{(x_n-x_0)(x_n-x_1)\dots(x_n-x_{n-1})} f(x_n)$$

Proof: Let $f(x_0), f(x_1), \dots, f(x_n)$ be $n+1$ corresponding values of the function $f(x)$ for $n+1$ different values $x_0, x_1, \dots, x_n$ (not necessarily equidistant) at the variable x .

Therefore the function $f(x)$ will be polynomial of degree n in x . Let this polynomial be $p_n(x)$.

$$F(x) = A_0(x-x_1)(x-x_2)\dots(x-x_n) + A_1(x-x_0)(x-x_2)\dots(x-x_n) + \dots + A_n(x-x_0)(x-x_1)\dots(x-x_{n-1}) = P_n(x) \quad (1)$$

Where $A_0, A_1, \dots, A_n$ are constants.

Since (1) is true for every value of x , so $P_n(x) = f(x_i)$, $i=0, 1, 2, \dots, n$, therefore the constants $A_0, A_1, \dots, A_n$ are obtained by substituting $x_0, x_1, \dots, x_n$ for x .

Substituting $x=x_0$ in (1), we obtain

$$F(x_0) = A_0(x_0-x_1)(x_0-x_2)\dots(x_0-x_n) + 0 + 0$$

$$A_0 = F(x_0) / (x_0-x_1)(x_0-x_2)\dots(x_0-x_n)$$

Similarly,

$$A_1 = F(x_1) / (x_1-x_0)(x_1-x_2)\dots(x_1-x_n)$$

Substituting the values of the constants $A_0, A_1, \dots, A_n$ in (1), the required formula is

$$F(x) = P_n(x) = \frac{(x-x_1)(x-x_2)\dots(x-x_n)}{(x_0-x_1)(x_0-x_2)\dots(x_0-x_n)} f(x_1) + \dots + \frac{(x-x_0)(x-x_1)\dots(x-x_{n-1})}{(x_n-x_0)(x_n-x_1)\dots(x_n-x_{n-1})} f(x_n)$$

Lagrange's formula is a relation in two variables x and y , in which any one can be taken as the independent variable, we can write a formula, where x will be a function of $f(y)$.

Therefore interchanging x and y , writing y as x and y_i as f_i , we obtain,

$$X = \frac{(f-f_1)(f-f_2)\dots(f-f_n)}{(f_0-f_1)(f_0-f_2)\dots(f_0-f_n)} (x_1) + \dots + \frac{(f-f_0)(f-f_1)\dots(f-f_{n-1})}{(f_n-f_0)(f_n-f_1)\dots(f_n-f_{n-1})} (x_n)$$

1. Use lagrange's formula to find $f(x)$ from the following data

X:	0	1	4	5
----	---	---	---	---

$$F(x): \quad 4 \quad 3 \quad 24 \quad 39$$

$$\text{Sol: } f(x) = \frac{(x-x_1)(x-x_2)(x-x_3)}{(x_0-x_1)(x_0-x_2)(x_0-x_3)} f(x_0) + \frac{(x-x_0)(x-x_2)(x-x_3)}{(x_1-x_0)(x_1-x_2)(x_1-x_3)} f(x_1) +$$

$$\frac{(x-x_0)(x-x_1)(x-x_3)}{(x_2-x_0)(x_2-x_1)(x_2-x_3)} f(x_2) + \frac{(x-x_0)(x-x_1)(x-x_2)}{(x_3-x_0)(x_3-x_1)(x_3-x_2)} f(x_3)$$

$$F(x) = \frac{(x-1)(x-4)(x-5)}{(0-1)(0-4)(0-5)} (4) + \frac{(x-0)(x-4)(x-5)}{(1-0)(1-4)(1-5)} (3) + \frac{(x-0)(x-1)(x-5)}{(4-0)(4-1)(4-5)} (24) + \frac{(x-0)(x-1)(x-4)}{(1-0)(1-4)(1-5)} (39)$$

$$F(x) = -\frac{1}{20} (x^3 - 10x^2 + 29x - 20)(4) + \frac{3}{12} x(x^2 - 9x + 20) - \frac{24}{12} x(x^2 - 6x + 5) +$$

$$\frac{39}{20} x(x^2 - 5x + 4)$$

$$F(x) = -\frac{1}{20} [(4 - 5 + 40 - 39)x^3 + (-40 + 45 - 240 + 195)x^2 + (116 - 100 + 200 - 156)x + (-80)]$$

$$F(x) = -\frac{1}{20} [0 - 40x^2 + 60x - 80]$$

$$F(x) = 2x^2 - 3x + 4$$

2. Use Lagrange's formula to interpolate the value of y at x=10 from the following data:

X:	5	6	9	11
Y:	12	13	14	16

Or interpolate the value of y at x=3 and x=10. Write the formula you are using. Why it?

Sol: Here $x_0=5, x_1=6, x_2=9, x_3=11$

$$Y_0=12, Y_1=13, Y_2=14, Y_3=16$$

So using Lagrange's interpolation formula, we get

$$F(3) = \frac{(3-6)(3-9)(3-11)}{(5-6)(5-9)(5-11)} (12) + \frac{(3-5)(3-9)(3-11)}{(6-5)(6-9)(6-11)} (13) + \frac{(3-5)(3-6)(3-11)}{(9-5)(9-6)(9-11)} (14) +$$

$$\frac{(3-5)(3-6)(3-9)}{(11-5)(11-6)(11-9)} (16)$$

$$F(3) = 7.2 + (-83.2) + 28 + (-9.6) = 7.2$$

So, now we have

$$x_0=3, x_1=5, x_2=6, x_3=9, x_4=11$$

$$Y_0=7.2, Y_1=12, Y_2=13, Y_3=14, Y_4=16$$

$$F(10) = \frac{(10-5)(10-6)(10-9)(10-11)}{(3-5)(3-6)(3-9)(3-11)}(7.2) + \frac{(10-3)(10-6)(10-9)(10-11)}{(5-3)(5-6)(5-9)(5-11)}(12) +$$

$$\frac{(10-3)(10-5)(10-9)(10-11)}{(6-3)(6-5)(6-9)(6-11)}(13) + \frac{(10-3)(10-5)(10-6)(10-11)}{(9-3)(9-5)(9-6)(9-11)}(14) +$$

$$\frac{(10-3)(10-5)(10-6)(10-9)}{(11-3)(11-5)(11-6)(11-9)}(16)$$

$$F(10) = -0.5 + 7 - 10.11 + 13.61 + 4.66$$

$$F(10) = 14.66$$

We use Lagrange's formula for less number of observations and greater accuracy.

References:

Advance Engineering Mathematics - H.K. Das, S. Chand

UNIT-5 (ADVANCE ENGINEERING MATHEMATICS)

Newton's Divided Difference Interpolation Formula

LECTURE NO: -39

Newton's Divided Difference Interpolation Formula

Interpolation is an estimation of a value within two known values in a sequence of values. Newton's divided difference interpolation formula is a interpolation technique used when the interval difference is not same for all sequence of values.

Suppose $f(x_0), f(x_1), f(x_2), \dots, f(x_n)$ be the $(n+1)$ values of the function $y=f(x)$ corresponding to the arguments $x=x_0, x_1, x_2, \dots, x_n$, where interval differences are not same

Then the first divided difference is given by

$$F(x_0, x_1) = \frac{f(x_1) - f(x_0)}{x_1 - x_0}$$

The second divided difference is given by

$$F(x_0, x_1, x_2) = \frac{f(x_1, x_2) - f(x_0, x_1)}{x_2 - x_0}$$

and so on...

Divided differences are symmetric with respect to the arguments i.e independent of the order of arguments.

so,

$$f[x_0, x_1] = f[x_1, x_0]$$

$$f[x_0, x_1, x_2] = f[x_2, x_1, x_0] = f[x_1, x_2, x_0]$$

By using first divided difference, second divided difference as so on .A table is formed which is called the divided difference table.

Divided difference table:

x_i	f_i	$F(x_i, x_j)$	$F(x_i, x_j, x_k)$
x1	f1		
		$f[x_1, x_2] = \frac{f_2 - f_1}{x_2 - x_1}$	
x2	f2		$f[x_1, x_2, x_3] = \frac{f[x_3, x_2] - f[x_2, x_1]}{x_3 - x_1}$
		$f[x_2, x_3] = \frac{f_3 - f_2}{x_3 - x_2}$	
x3	f3		

Examples:

Input : Value at 7

x	5	6	9	11
y=f(x)	12	13	14	16

Output :

x	y=f(x)	f[xi,xj]	f[xi,xj,xk]	f[xi,xj,xk,xl]
5	12			
		1		
6	13		$-\frac{1}{6}$	
		$\frac{1}{3}$		$\frac{1}{20}$
9	14		$\frac{2}{15}$	
		1		
11	16			

Value at 7 is 13.47

The divided difference $f[x_0, x_1, x_2, \dots, x_n]$, sometimes also denoted $[x_0, x_1, x_2, \dots, x_n]$ (Abramowitz and Stegun 1972), on $n+1$ points $x_0, x_1, \dots, x_n$ of a function $f(x)$ is defined by $f[x_0] \equiv f(x_0)$ and

$$f[x_0, x_1, \dots, x_n] = \frac{f[x_0, \dots, x_{n-1}] - f[x_1, \dots, x_n]}{x_0 - x_n} \quad (1)$$

for $n \geq 1$. The first few differences are

$$f[x_0, x_1] = \frac{f_0 - f_1}{x_0 - x_1} \quad (2)$$

$$f[x_0, x_1, x_2] = \frac{f[x_0, x_1] - f[x_1, x_2]}{x_0 - x_2} \quad (3)$$

$$f[x_0, x_1, \dots, x_n] = \frac{f[x_0, \dots, x_{n-1}] - f[x_1, \dots, x_n]}{x_0 - x_n}. \quad (4)$$

Defining

$$\pi_n(x) \equiv (x - x_0)(x - x_1) \cdots (x - x_n) \quad (5)$$

and taking the derivative

$$\pi'_n(x_k) = (x_k - x_0) \cdots (x_k - x_{k-1})(x_k - x_{k+1}) \cdots (x_k - x_n) \quad (6)$$

gives the identity

$$f[x_0, x_1, \dots, x_n] = \sum_{k=0}^n \frac{f_k}{\pi'_n(x_k)}. \quad (7)$$

Consider the following question: does the property

Newton's Divided Difference Interpolation Formula: Newton's Divided Difference is a way of finding an interpolation polynomial (a polynomial that fits a particular set of points or data). Similar to Lagrange's method for finding an interpolation polynomial, it finds the same interpolation polynomial due to the uniqueness of interpolation polynomials. Newton's Divided Difference uses the following equation called the divided difference to accomplish this task:

$$\begin{aligned} & f(x_0, x_1, x_2, \dots, x_n) \\ &= \frac{f(x_1, x_2, x_3, \dots, x_n) - f(x_0, x_1, x_2, \dots, x_{n-1})}{x_n - x_0} \end{aligned}$$

$$F(x) = f(x_0) + (x - x_0)f(x_0, x_1) + (x - x_0)(x - x_1)f(x_0, x_1, x_2) + \dots + (x - x_0)(x - x_1) \dots$$

$$(x - x_n) f(x_0, x_1, x_2, \dots, x_n)$$

Example : Compute **f(0.3)** for the data

x	0	1	3	4	7
f	1	3	49	129	813

using Newton's divided difference formula.

Solution : Divided difference table

x_i	f_i			
0	1			
		2		
1	3		7	
		23		3
3	49		19	
		80		3
4	129		37	
		228		
7	813			

Now Newton's divided difference formula is

$$f(x) = f[x_0] + (x - x_0) f[x_0, x_1] + (x - x_0)(x - x_1) f[x_0, x_1, x_2] + (x - x_0)(x - x_1)(x - x_2) f[x_0, x_1, x_2, x_3]$$

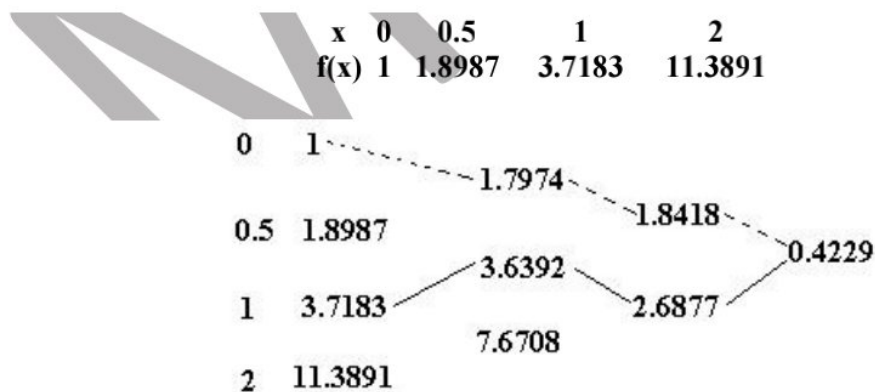
$$f(0.3) = 1 + (0.3 - 0) 2 + (0.3)(0.3 - 1) 7 + (0.3)(0.3 - 1)(0.3 - 3) 3$$

$$= 1.831$$

Since the given data is for the polynomial $f(x) = 3x^3 - 5x^2 + 4x + 1$ the analytical value is $f(0.3) = 1.831$

The analytical value is matched with the computed value because the given data is for a third degree polynomial and there are five data points available using which one can approximate any data exactly upto fourth degree polynomial.

Example : Find $f(1.5)$ from the data points



$f(1.5)$ along the dotted line is

$$f(1.5) = 1 + 1.5 \times 1.7974 + 1.5(1) \times 1.8418 + (1.5)(1)(0.5) \times 0.4229$$

$$= 6.770$$

Similarly **f(1.5)** along the solid line is

$$\mathbf{f(1.5) = 3.7183 + (1.5 - 1) \times 3.6392 + (1.5 - 1)(1.5 - 0.5) \times 2.6877 + (1.5 - 1)(1.5 - 0.5)(1.5 - 2) \times 0.4229 = 6.770}$$

The data is given for **f(x) = x² + e^x** and the analytical value for **f(1.5) = 6.7317**

References:

Advance Engineering Mathematics - H.K.Das, S. Chand

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UNIT-5 (ADVANCE ENGINEERING MATHEMATICS)

NUMERICAL DIFFERENTIATION

LECTURE NO: -40

Numerical Differentiation

In the case of differentiation, we first write the interpolating formula on the interval (x_0, x_n) and then differentiate the polynomial to get the derivative of the function. When the tabular points are equidistant, one uses either the Newton's Forward/Backward; otherwise Lagrange's formula is used. Newton's Forward/Backward formula is used depending upon the location of the point at which the derivative is to be computed. In case the given point is near the mid-point of the interval, Stirling's formulae's can be used. Other formulae are also used but Stirling's formulae's is best in all, so we use it when central values are required in differentiation. If forward and backward formula are used then we obtain other values than central value. Formulae's of Newton forward and backward formulae's are used when we require the values near starting and ending values.

Numerical differentiation is applied all difference formulae's.

Recall, that the Newton's forward interpolating polynomial is given by

$$f(x) = f(x_0 + hu) \approx y_0 + \Delta y_0 u + \frac{\Delta^2 y_0}{2!} (u(u-1)) + \cdots + \frac{\Delta^k y_0}{k!} \{u(u-1) \cdots (u-k+1)\} \\ + \cdots + \frac{\Delta^n y_0}{n!} \{u(u-1) \cdots (u-n+1)\}.$$

Differentiating, we get the approximate value of the first derivative at x as

$$\frac{df}{dx} = \frac{1}{h} \frac{df}{du} \approx \frac{1}{h} \left[\Delta y_0 + \frac{\Delta^2 y_0}{2!} (2u-1) + \frac{\Delta^3 y_0}{3!} (3u^2 - 6u + 2) + \cdots \right. \\ \left. + \frac{\Delta^n y_0}{n!} \left(nu^{n-1} - \frac{n(n-1)^2}{2} u^{n-2} + \cdots + (-1)^{(n-1)} (n-1)! \right) \right].$$

where, $u = \frac{x-x_0}{h}$

Thus, an approximation to the value of first derivative at $x=x_0$ i.e. $u=0$ is obtained as :

$$\left. \frac{df}{dx} \right|_{x=x_0} = \frac{1}{h} \left[\Delta y_0 - \frac{\Delta^2 y_0}{2} + \frac{\Delta^3 y_0}{3} - \dots + (-1)^{(n-1)} \frac{\Delta^n y_0}{n} \right].$$

Similarly, using Stirling's formula

$$\begin{aligned} f(x_0^* + hu) &\approx y_0^* + u \frac{\Delta y_{-1}^* + \Delta y_0^*}{2} + u^2 \frac{\Delta^2 y_{-1}^*}{2!} + \frac{u(u^2 - 1)}{2} \frac{\Delta^3 y_{-2}^* + \Delta^3 y_{-1}^*}{3!} \\ &\quad + u^2(u^2 - 1) \frac{\Delta^4 y_{-2}^*}{4!} + \frac{u(u^2 - 1)(u^2 - 2^2)}{2} \frac{\Delta^5 y_{-3}^* + \Delta^5 y_{-2}^*}{5!} + \dots \end{aligned}$$

Therefore

$$\begin{aligned} \frac{df}{dx} &= \frac{1}{h} \frac{df}{du} \approx \frac{1}{h} \left[\frac{\Delta y_{-1}^* + \Delta y_0^*}{2} + u \Delta^2 y_{-1}^* + \frac{(3u^2 - 1)}{2} \times \frac{(\Delta^3 y_{-2}^* + \Delta^3 y_{-1}^*)}{3!} \right. \\ &\quad \left. + 2u(2u^2 - 1) \frac{\Delta^4 y_{-2}^*}{4!} + \frac{(5u^4 - 15u^2 + 4)(\Delta^5 y_{-3}^* + \Delta^5 y_{-2}^*)}{2 \times 5!} + \dots \right] \end{aligned}$$

Thus, the derivative at $x=x_0^*$ is obtained as:

$$\left. \frac{df}{dx} \right|_{x=x_0^*} = \frac{1}{h} \left[\frac{\Delta y_{-1}^* + \Delta y_0^*}{2} - \frac{(1)}{2} \times \frac{(\Delta^3 y_{-2}^* + \Delta^3 y_{-1}^*)}{3!} + \frac{4 \times (\Delta^5 y_{-3}^* + \Delta^5 y_{-2}^*)}{2 \times 5!} + \dots \right].$$

Numerical differentiation using Stirling's formula is found to be more accurate than that with the Newton's difference formulae. Also it is more convenient to use.

Now higher derivatives can be found by successively differentiating the interpolating polynomials. Thus e.g. using, we get the second derivative at $x=x_0$ as

$$\left. \frac{d^2 f}{dx^2} \right|_{x=x_0} = \frac{1}{h^2} \left[\Delta^2 y_0 - \Delta^3 y_0 + \frac{2 \times 11 \times \Delta^4 y_0}{4!} - \dots \right].$$

EXAMPLE 13.2.2 Compute from following table the value of the derivative of $y=f(x)$ at $x = 1.7489$:

x	1.73	1.74	1.75	1.76	1.77
y	1.772844100	1.155204006	1.737739435	1.720448638	1.703329888

Solution: We note here that

$$x_0 = 1.75, h = 0.01,$$

$$u = (1.7489 - 1.75)/0.01 = -0.11$$

so

$$\Delta y_0 = -0.0017290797, \Delta^2 y_0 = 0.0000172047, \Delta^3 y_0 = -0.0000001712,$$

and

$$\Delta y_{-1} = -0.0017464571, \Delta^2 y_{-1} = 0.0000173774, \Delta^3 y_{-1} = -0.0000001727,$$

$$\Delta^3 y_{-2} = -0.0000001749, \Delta^4 y_{-2} = -0.0000000022$$

Thus, $f(1.7489)$ is obtained as:

(i) Using Newton's Forward difference formula,

$$f'(1.4978) \approx \frac{1}{0.01} \left[-0.0017290797 + (2 \times -0.11 - 1) \times \frac{0.0000172047}{2} \right. \\ \left. + (3 \times (-0.11)^2 - 6 \times -0.11 + 2) \times \frac{-0.0000001712}{3!} \right] = -0.173965150143.$$

Using Stirling's formula, we get:

$$f'(1.4978) \approx \frac{1}{.01} \left[\frac{(-0.0017464571) + (-0.0017290797)}{2} + (-0.11) \times 0.0000173774 \right. \\ \left. + \frac{(3 \times (-0.11)^2 - 1) ((-0.0000001749) + (-0.0000001727))}{2 \cdot 3!} \right. \\ \left. + 2 \times (-0.11) \times (2(-0.11)^2 - 1) \times \frac{(-0.0000000022)}{4!} \right] \\ = -0.17396520185$$

It may be pointed out here that the above table is for $f(x)=e^{-x}$, whose derivative has the value -0.1739652000 at $x = 1.7489$.

References:

Advance Engineering Mathematics - H.K.Das, S. Chand

UNIT-5 (ADVANCE ENGINEERING MATHEMATICS)

NUMERICAL INTEGRATION

LECTURE NO: -41

Numerical Integration

There are two topics with similar names:

- Reverse of differentiation
- Indefinite integral

$$\int_a^b f(x)dx$$

= most general antiderivative for $f(x)$

$f(x)dx$ (read integral from a to b of the function f).

- Definite integral

This is related to summation (it is a limit of sums of a certain kind). The integral sign was originally invented as a modified S (for sum).

There is no reason to expect a connection between these two different things, but there is.

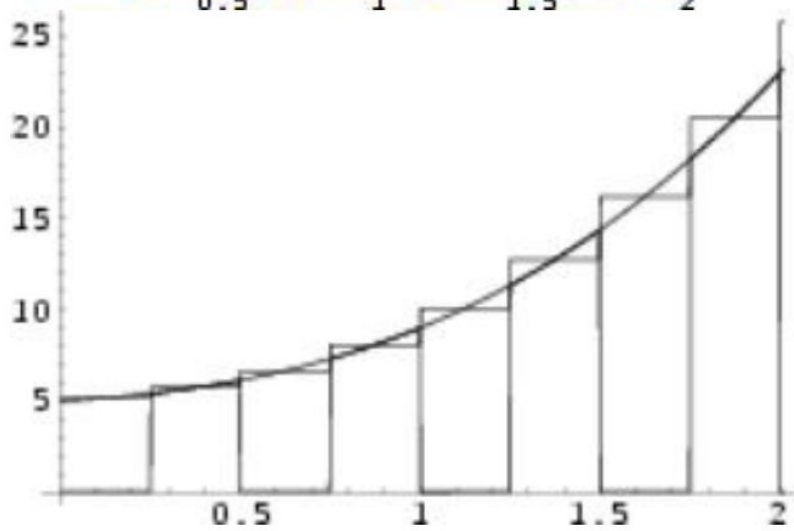
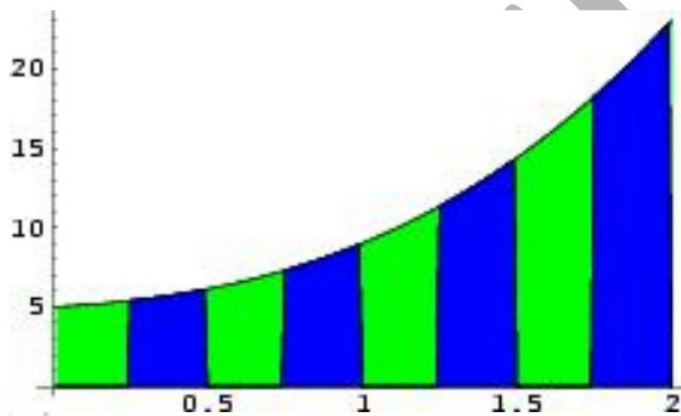
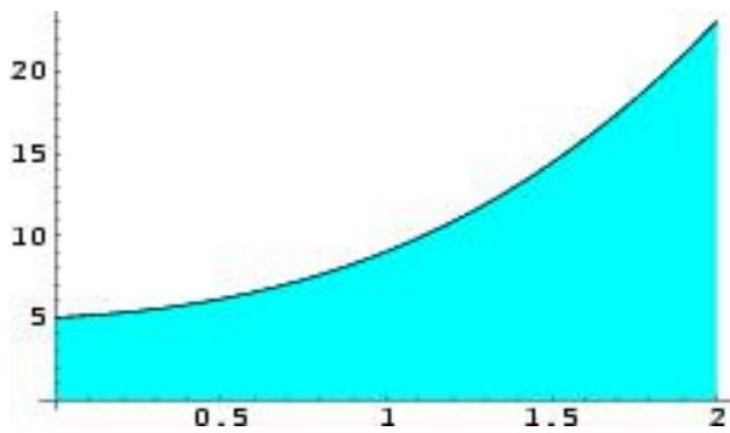
What we need is the idea of a Riemann sum.

$$\int_a^b f(x)dx$$

Graphical interpretation:

graph $y = f(x)$, the x -axis and the vertical lines $x = a$, $x = b$.

Problems: need a way to compute the area of such a shape.



Notation. We need notation to write down formulae for these pictures.

$$\int_a^b f(x)dx$$

n subdivisions of interval $[a, b]$

Number division point $a = x_0 < x_1 < x_2 < \dots < x_n = b$

Heights of rectangles $y_j = f(x_j^*)$, $x_{j-1} \leq x_j^* \leq x_j$ any point.

"Area" of j th rectangle = width $\times$ height = $(x_j - x_{j-1})y_j$

Total "area" = sum of these

$$= y_1(x_1 - x_0) + y_2(x_2 - x_1) + \dots + y_n(x_n - x_{n-1})$$

$$= f(x_1^*)(x_1 - x_0) + f(x_2^*)(x_2 - x_1) + \dots + f(x_n^*)(x_n - x_{n-1})$$

(called *Riemann sums* for the integral)

Numerical integration is the process in which we evaluate a definite integral from a set of tabulated value of a implicit function $f(x)$.

Consider a set of data $(x_0, y_0), (x_1, y_1), \dots, (x_n, y_n)$ of a function $y=f(x)$, where $f(x)$ is not known analytically. We wish to find the value of the definite integral

$$\int_a^b f(x)dx \dots \dots \dots (1)$$

For this we divide the interval $[a, b]$ in to n subintervals of width h so that

$$x_0=a, x_1=x_0+h, x_2=x_0+2h, \dots, x_n=x_0+nh$$

Now

$$I = \int_a^b f(x)dx = \int_{x_0}^{x_0+nh} y dx \dots \dots \dots (2)$$

To evaluate (2), we approximate y by Newton forward formula

$$I = \int_{x_0}^{x_0+nh} \left[y_0 + u\Delta y_0 + \frac{u(u-1)}{2}\Delta^2 y_0 + \dots \right] dx$$

$$u = (x - x_0)/h$$

$$x = x_0 + uh$$

$$\text{and at } x = x_0; u=0$$

$$\text{and at } x = x_0 + nh; u=n$$

$$\text{and if } x = x_0 + uh \text{ then } dx = hdu$$

so $I = \int_0^n [y_0 + u\Delta y_0 + \frac{u(u-1)}{2}\Delta^2 y_0 + \dots] dx$

integrating w.r.t. u , we have

$$I = h[u y_0 + u^2 \frac{\Delta y_0}{2!} + \left(\frac{u^3}{3} - \frac{u^2}{2}\right) \frac{\Delta^2 y_0}{2!} + \frac{1}{3!} \left(\frac{u^4}{4} - u^3 + u^2\right) \Delta^3 y_0 + \dots]$$

$$\int_{x_0}^{x_0+nh} y dx = h[n y_0 + n^2 \frac{\Delta y_0}{2} + \left(\frac{n^3}{3} - \frac{n^2}{2}\right) \frac{\Delta^2 y_0}{2} + \frac{1}{6} \left(\frac{n^4}{4} - n^3 + n^2\right) \Delta^3 y_0 + \dots] \dots \dots \dots (3)$$

Equation (3) is known as Newton-cotes quadrature formula.

From this formula we obtain the three rules:

1. Trapezoidal rule
 2. Simpson's 1/3 rule
 3. Simpson's 3/8 rule
- Putting $n=1$, $n=2$ and $n=3$.

Trapezoidal Rule

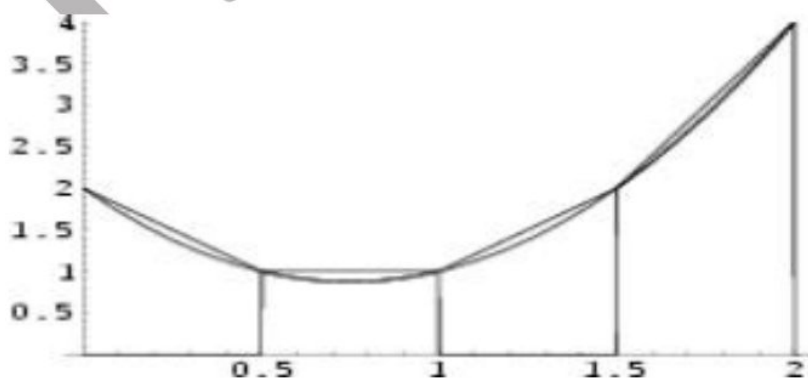
The *trapezoidal rule* is a technique for finding definite integrals

$$\int_a^b f(x) dx$$

numerically.

It is one step more clever than using Riemann sums. In Riemann sums, what we essentially do is approximate the graph $y = f(x)$ by a step graph and integrate the step graph.

In the Trapezoidal rule, we approximate $y = f(x)$ by a continuous graph made up of bits of line



Use n equal divisions. $h = (b-a)/n$. Let $y_i = f(x_i)$ (for $0 \leq i \leq n$).

The trapezoidal rule formula can be
 $\int_a^b f(x) dx = h/2 (y_0 + 2(y_1 + y_2 + y_3 + \dots + y_{n-1}) + y_n)$

Proof (of this formula) uses area of i th trapezoid = $\int_a^b f(x) dx = h/2 (y_0 + 2(y_1 + y_2 + y_3 + \dots + y_{i-1}) + y_i)$

Simpsons Rule. Idea: For

$$\int_a^b f(x) dx$$

choose n **even** and preferably large. Divide the

interval $[a, b]$ into n equal sections each of width $h = (b-a)/n$ division points
 $x = a + ih$

($0 \leq i \leq n$), corresponding values $y_i = f(x_i)$. Pair off intervals and approximate $y = f(x)$ on each pair by a quadratic graph.

$\int_a^b f(x) dx \approx$ sum of integrals of these quadratics. The formula we get out of this is

$$\int_a^b f(x) dx = h/3 (y_0 + 4(y_1 + y_3 + y_5 + \dots + y_{n-1}) + 2(y_2 + y_4 + y_6 + \dots + y_{n-2}) + y_n)$$

Formula

1. Trapezoidal Rule

$$\int y dx = h/2 (y_0 + 2(y_1 + y_2 + y_3 + \dots + y_{n-1}) + y_n)$$

2. Simpsons 13 Rule

$$\int y dx = h/3 (y_0 + 4(y_1 + y_3 + y_5 + \dots + y_{n-1}) + 2(y_2 + y_4 + y_6 + \dots + y_{n-2}) + y_n)$$

3. Simpsons 38 Rule

$$\int y dx = 3h/8 (y_0 + 2(y_3 + y_6 + \dots + y_{n-3}) + 3(y_1 + y_2 + y_4 + y_5 + \dots + y_{n-2} + y_{n-1}) + y_n)$$

Examples

1. Find Solution using all method

x	f(x)
0.0	1.0000
0.1	0.9975
0.2	0.9900
0.3	0.9776
0.4	0.8604

Solution:

The value of table for X and Y

X	0 x0=0	0.1 x1=0.1	0.2 x2=0.2	0.3 x3=0.3	0.4 x4=0.4
Y	1 y0=1	0.9975 y1=0.9975	0.99 y2=0.99	0.9776 y3=0.9776	0.8604 y4=0.8604

Using Trapezoidal Rule

$$\int Y dx = h[2(y_0 + y_4) + 2(y_1 + y_2 + y_3)]$$

$$\int Y dx = 0.12[2(1 + 0.8604) + 2(0.9975 + 0.99 + 0.9776)]$$

$$\int Y dx = 0.12[2(1.8604) + 2(2.9651)]$$

$$\int Y dx = 0.38953$$

Solution By Trapezoidal Rule is 0.38953

Using Simpsons 13 Rule

$$\int Y dx = h[3(y_0 + y_4) + 4(y_1 + y_3) + 2(y_2)]$$

$$\int Y dx = 0.12[3(1 + 0.8604) + 4(0.9975 + 0.9776) + 2(0.99)]$$

$$\int Y dx = 0.12[3(1.8604) + 4(1.9751) + 2(0.99)]$$

$$\int Y dx = 0.39136$$

Solution By Simpson's 13 Rule is 0.39136

Using Simpson's 38 Rule

$$\int Y dx = \frac{3h}{8}[y_0 + 3(y_1 + y_3) + 4(y_2) + y_4]$$

$$\int Y dx = \frac{3 \times 0.12}{8}[1 + 3(0.9975 + 0.9776) + 4(0.99) + 0.8604]$$

$$\int Y dx = \frac{3 \times 0.12}{8}[1 + 3(1.9751) + 4(0.99) + 0.8604]$$

$$\int Y dx = 0.36668$$

Solution By Simpson's 3/8 Rule is 0.36668

UNIT-5 (ADVANCE ENGINEERING MATHEMATICS)

NUMERICAL SOLUTION OF ORDINARY DIFFERENTIAL EQUATIONS- EULER'S METHOD

LECTURE NO: -42

Numerical Solution of Ordinary Differential Equations-Euler's Method:

Though in principle it is possible to use Taylor's method of any order for the given initial value problem to get good approximations, it has few drawbacks like

- The scheme assumes the existence of all higher order derivatives for the given function $f(x, y)$ which is not a requirement for the existence of the solution for any first order initial value problem.
- Even the existence of these higher derivatives is guaranteed it may not be easy to compute them for any given $f(x, y)$.
- Because of the usage of higher order derivatives in the formula it is not convenient to write computer programs, that is the method is more suited for hand calculations.

To overcome these difficulties, Euler developed a scheme by approximating y' in the given ivp. The scheme is as follows:

The derivative term in the first order ivp

$$y' = f(x, y), \quad y(x_0) = y_0$$

is approximated by making use of Taylor series approximation of the dependent variable $y(x)$ at the point x_{i+1} . That is

$$\begin{aligned} y(x_{i+1}) &= y(x_i + Dx) = y(x_i) + Dx y'(x_i) + \frac{(Dx)^2}{2} y''(x_i) + \dots \\ &= y(x_i) + Dx f(x_i, y_i) + \frac{(Dx)^2}{2} y''(x_i) + \dots \end{aligned}$$

$$(\dots y'(x_i) = f(x_i, y_i))$$

That is,

$$\text{for } i = 0, \quad y_1 = y_0 + Dx f_0$$

$$i = 1, \quad y_2 = y_1 + Dx f_1$$

!

$$i = n-1, \quad y_n = y_{n-1} + Dx f_{n-1}$$

Since y_0 and hence f_0 are known (from initial condition) in the equation corresponding to $i = 0$, all the terms on the r.h.s are known. So y_1 that is, y at x_1 is calculated easily from this equation. Similarly once y_1 is known, r.h.s of the equation corresponding to $i = 1$ is also known so y_2 can be computed. As we proceed in the same way until $i = n-1$, y_n can be obtained. This is an explicit method because in any equation there is only one unknown which can be separated to the left side of the equation.

This is very simple method to use because simple steps are used to calculate values of y . but it not used because approximate value is not near accuracy.

Example 1:

Find $y(0.5)$ if y is the solution of IVP $y' = -2x - y$, $y(0) = -1$ using Euler's method with step length 0.1. Also find the error in the approximation.

Solution: $f(x, y) = -2x - y$,

$$y_1 = y_0 + h f(x_0, y_0) = -1 + 0.1 * (-2*0 - (-1)) = -0.8999$$

$$y_2 = y_1 + h f(x_1, y_1) = -0.8999 + 0.1 * (-2*0.1 - (-0.8999)) = -0.8299$$

$$y_3 = y_2 + h f(x_2, y_2) = -0.8299 + 0.1 * (-2*0.2 - (-0.8299)) = -0.7869$$

$$y_4 = y_3 + h f(x_3, y_3) = -0.7869 + 0.1 * (-2*0.3 - (-0.7869)) = -0.7683$$

$$y_5 = y_4 + h f(x_4, y_4) = -0.7683 + 0.1 * (-2*0.4 - (-0.7683)) = -0.7715$$

Example 2:

Solve the differential equation $y' = x/y$, $y(0)=1$ by Euler's method to get $y(1)$. Use the step lengths $h = 0.1$ and 0.2 and compare the results with the analytical solution ($y^2 = 1 + x^2$)

Solution: $f(x, y) = x/y$,

with $h = 0.1$

$$y_1 = y_0 + h f(x_0, y_0) = 1.0 + 0.1*0.0/1.0 = 1.00$$

$$y_2 = y_1 + h f(x_1, y_1) = 1.0 + 0.1*0.1/1.0 = 1.01$$

$$y_3 = y_2 + h f(x_2, y_2) = 1.01 + 0.1*0.2/1.01 = 1.0298$$

$$y_4 = y_3 + h f(x_3, y_3) = 1.0298 + 0.1*0.3/1.0298 = 1.0589$$

$$y_5 = y_4 + h f(x_4, y_4) = 1.0589 + 0.1*0.4/1.0589 = 1.0967$$

$$y_6 = y_5 + h f(x_5, y_5) = 1.0967 + 0.1*0.5/1.0967 = 1.1423$$

$$y_7 = y_6 + h f(x_6, y_6) = 1.1423 + 0.1*0.6/1.1423 = 1.1948$$

$$y_8 = y_7 + h f(x_7, y_7) = 1.1948 + 0.1*0.7/1.1948 = 1.2534$$

$$y_9 = y_8 + h f(x_8, y_8) = 1.2534 + 0.1*0.8/1.2534 = 1.3172$$

$$y_{10} = y_9 + h f(x_9, y_9) = 1.3172 + 0.1*0.9/1.3172 = 1.3855$$

with $h = 0.2$

$$y_1 = y_0 + h f(x_0, y_0) = 1.0 + 0.2*0.0/1.0 = 1.0$$

$$y_2 = y_1 + h f(x_1, y_1) = 1.0 + 0.2*0.2/1.0 = 1.0400$$

$$y_3 = y_2 + h f(x_2, y_2) = 1.0400 + 0.2*0.4/1.0400 = 1.1169$$

$$y_4 = y_3 + h f(x_3, y_3) = 1.1169 + 0.2*0.6/1.1169 = 1.2243$$

$$y_5 = y_4 + h f(x_4, y_4) = 1.2243 + 0.2*0.8/1.2243 = 1.3550$$

Comparison of numerical and analytical solutions:

x	Numerical Solution		Analytical solution
	h = 0.1	h = 0.2	
0.0	1.0	1.0	1.0
0.1	1.0		1.0050
0.2	1.01	1.0	1.0198

0.3	1.0298		1.0440
0.4	1.0589	1.0400	1.0770
0.5	1.0967		1.1180
0.6	1.1423	1.1169	1.1662
0.7	1.1948		1.2207
0.8	1.2534	1.2243	1.2806
0.9	1.3172		1.3454
1.0	1.3855	1.3550	1.4142

UNIT-5 (ADVANCE ENGINEERING MATHEMATICS)

MODIFIED EULER'S METHOD

LECTURE NO: -43

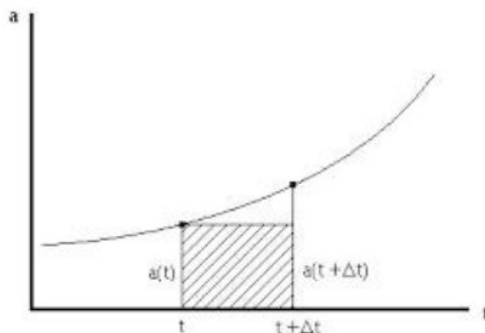
Modified Euler's Method

If we are given a first order differential equation that could be solved. There are many different problems that are very difficult to solve. There are different categories of first order equations like linear equation, separable equation, nonlinear equation, exact equation, homogeneous equation, Bernoulli equation, and non-homogeneous equations. So we use another method called Euler's Method. In Euler's Method a differential equation given for which we found the slope of a function and take a step size for the integral (if step size is smaller then accuracy in approximate value we will get.). we take another new point for new approximate value and add this to old value and continue the process. At last we near approximate value to root. The formula for this is,

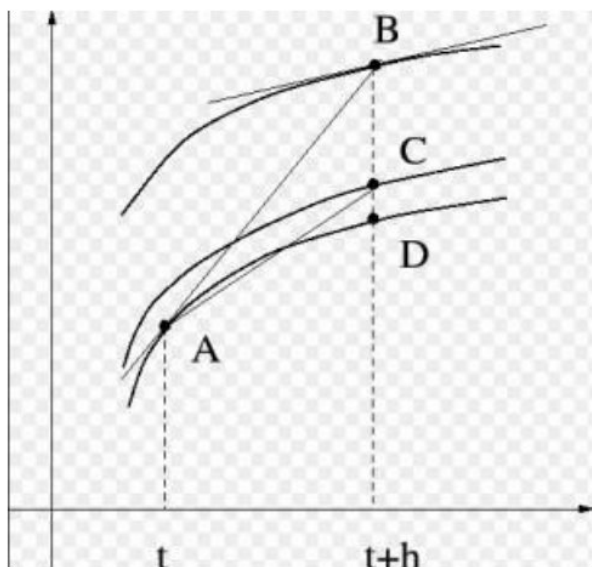
$$y_{n+1} = y_n + hf(t_n, y_n)$$

However, the error for the Euler's Method depends on the step size. the only way to decrease the error is to reduce the step size, but it will increase the amount of calculations. it only roughly decreases the error by half.

Now, we introduce an improved Euler's Method. This method is quite similar to the Euler's method. In the Euler's Method we approximate the function by a rectangular shape (see graph below):



However, this approximate does not include the area that under the curve. To improve the approximation, we use the improved Euler's method. The improved method, we use the average of the values at the initially given point and the new point. We define the integral with a trapezoid instead of a rectangle. The trapezoid has more area covered than the rectangle area. It will also provide a more accurate approximation.



It is hard to predict the solution curve is concave up or concave down in reality. The ideal prediction line would exactly hit the curve at next predict point. The Euler's Method generates the slope based on the initial point, and we don't know if the next point will be on this slope line, unless we use a computer to plot the equation. Sometimes, we might overestimate the value or underestimate the value. The Improved Euler's Method addressed these problems by finding the average of the slope based on the initial point and the slope of the new point, which will give an average point to estimate the value. It also decreases the errors that Euler's Method would have.

The improved Euler's Method simply divided into three steps as following:

Steps in Improved Euler's Method:

Step 1 find the $k_1 = f(t_n, y_n)$

Step 2 find the $k_2 = f(t_n + h, y_n + hk_1)$

Step 3: find $y(n+1) = f(t_n + h, y_n + h((k_1 + k_2)/2))$

Given a first order linear equation $y' = t^2 + 2y$, $y(0) = 1$, estimate $y(2)$, step size is 0.5

Point 1:

$$(t_n, y_n) = (0, 1)$$

$$t_{n+1} = 0 + 0.5 = 0.5$$

$$1. \quad k_1 = f(t_n, y_n) = f(0, 1) = 0^2 + 2 \times 1 = 2$$

$$2. \quad k_2 = f(t_n + h, y_n + hk_1) = f(0.5, 1 + 0.5 \times 1.2)$$

$$f(0.5, 1.6) = 0.5^2 + 2 \times 1.6 = 3.45$$

$$3. \quad y_{n+1} = y_n + h \cdot \frac{k_1 + k_2}{2} = 1 + 0.5 \times \frac{2 + 3.45}{2} = 2.3625$$

Point 2:

$$(t_n, y_n) = (0.5, 2.3625)$$

$$t_{n+1} = 0.5 + 0.5 = 1$$

$$4. \quad k_1 = f(t_n, y_n) = f(0.5, 2.3625) = 0.5^2 + 2 \times 2.3625 = 4.975$$

$$5. \quad k_2 = f(t_n + h, y_n + hk_1) = f(1, 2.3625 + 0.5 \times 4.975)$$

$$f(1, 4.85) = 1^2 + 2 \times 4.85 = 10.7$$

$$6. \quad y_{n+1} = y_n + h \cdot \frac{k_1 + k_2}{2} = 2.3625 + 0.5 \times \frac{4.975 + 10.7}{2} = 6.28125$$

Point 3:

$$(t_n, y_n) = (1, 6.28125)$$

$$t_{n+1} = 1 + 0.5 = 1.5$$

$$7. \quad k_1 = f(t_n, y_n) = f(1, 6.28125) = 1^2 + 2 \times 6.28125 = 13.5625$$

$$8. \quad k_2 = f(t_n + h, y_n + hk_1) = f(1.5, 6.28125 + 0.5 \times 13.5625) \\ f(1.5, 13.0625) = 1.5^2 + 2 \times 13.0625 = 28.375$$

$$9. \quad y_{n+1} = y_n + h \cdot \frac{k_1 + k_2}{2} = 6.28125 + 0.5 \times \frac{13.5625 + 28.375}{2} = 16.765625$$

Point 4:

$$(t_n, y_n) = (1.5, 16.765625)$$

$$t_{n+1} = 1.5 + 0.5 = 2.0$$

$$10. \quad k_1 = f(t_n, y_n) = f(1.5, 16.765625) = 1.5^2 + 2 \times 16.765625 = 35.78125$$

$$11. \quad k_2 = f(t_n + h, y_n + hk_1) = f(2.0, 16.765625 + 0.5 \times 35.78125) \\ f(2.0, 34.65625) = 2.0^2 + 2 \times 34.65625 = 73.3125$$

$$12. \quad y_{n+1} = y_n + h \cdot \frac{k_1 + k_2}{2} = 16.765625 + 0.5 \times \frac{35.78125 + 73.3125}{2} = \mathbf{44.0390625}$$

The value for $y(2.0) = 44.0390625$

UNIT-5 (ADVANCE ENGINEERING MATHEMATICS)

RUNGE-KUTTA 4TH ORDER METHOD

LECTURE NO: -44

Runge-Kutta 4th Order Method

Given following inputs,

- An ordinary differential equation that defines value of dy/dx in the form x and y .
- Initial value of y , i.e., $y(0)$

Thus we are given below.

$$dy/dx = f(x, y)$$

$$y(0) = y_0$$

The task is to find value of unknown function y at a given point x .

The Runge-Kutta method finds approximate value of y for a given x . Only first order ordinary differential equations can be solved by using the Runge Kutta 4th order method

Below is the formula used to compute next value y_{n+1} from previous value y_n . The value of n are $0, 1, 2, 3, \dots, (x - x_0)/h$. Here **h is step height** and **$x_{n+1} = x_n + h$**

$$K_1 = hf(x_n, y_n)$$

$$K_2 = hf(x_n + h/2, y_n + k_1/2)$$

$$K_3 = hf(x_n + h/2, y_n + k_2/2)$$

$$K_4 = hf(x_n + h, y_n + k_3)$$

$$Y_{n+1} = y_n + 1/6(k_1 + 2k_2 + 2k_3 + k_4)$$

The formula basically computes next value y_{n+1} using current y_n plus weighted average of four increments.

- k_1 is the increment based on the slope at the beginning of the interval, using y
- k_2 is the increment based on the slope at the midpoint of the interval, using $y + hk_1/2$.
- k_3 is again the increment based on the slope at the midpoint, using $y + hk_2/2$.
- k_4 is the increment based on the slope at the end of the interval, using $y + hk_3$.

Example

- Use Runge-Kutta Method of Order 4 to solve the following, using a step size of $h=0.1$ for $0 \leq x \leq 1$
- $dy/dx = \frac{5x^2 - y}{e^{x+y}}$, $y(0)=1$

Let us look at an example:

$$y' = y - t^2 + 1$$

$$y(0) = 0.5$$

The exact solution for this problem is $y = t^2 + 2t + \frac{1}{2} - \frac{1}{2}e^t$, and we are interested in the value of y for $0 \leq t \leq 2$.

1. We first solve this problem using RK4 with $h = 0.5$. From $t = 0$ to $t = 2$ with step size $h = 0.5$, it takes 4 steps: $t_0 = 0$, $t_1 = 0.5$, $t_2 = 1$, $t_3 = 1.5$, $t_4 = 2$.

Step 0 $t_0 = 0$, $w_0 = 0.5$.

Step 1 $t_1 = 0.5$

$$k_1 = hf(t_0, w_0) = 0.5f(0, 0.5) = 0.75$$

$$k_2 = hf(t_0 + h/2, w_0 + k_1/2) = 0.5f(0.25, 0.875) = 0.90625$$

$$K_3 = hf(t_0 + h/2, w_0 + k_2/2) = 0.5f(0.25, 0.953125) = 0.9453125$$

$$K_4 = hf(t_0 + h, w_0 + K_3) = 0.5f(0.5, 1.4453125) = 1.09765625$$

$$w_1 = w_0 + (k_1 + 2k_2 + 2K_3 + K_4)/6 = 1.425130208333333$$

Step 2 $t_2 = 1$

$$k_1 = hf(t_1, w_1) = 0.5f(0.5, 1.425130208333333) = 1.087565104166667$$

$$k_2 = hf(t_1 + h/2, w_1 + k_1/2) =$$

$$K_4 = hf(t_1 + h, w_1 + K_3) = 0.5f(1, 2.657246907552083) = 1.328623453776042$$

$$w_2 = w_1 + (k_1 + 2k_2 + 2K_3 + K_4)/6 = 2.639602661132812$$

Step 3 $t_3 = 1.5$

$$k_1 = hf(t_2, w_2) = 0.5f(1, 2.639602661132812) = 1.319801330566406$$

$$k_2 = hf(t_2 + h/2, w_2 + k_1/2) = 0.5f(1.25, 3.299503326416016) = 1.368501663208008$$

$$K_3 = hf(t_2 + h/2, w_2 + k_2/2) = 0.5f(1.25, 3.323853492736816) = 1.380676746368408$$

$$K_4 = hf(t_2 + h, w_2 + K_3) = 0.5f(1.5, 4.020279407501221) = 1.385139703750610$$

$$w_3 = w_2 + (k_1 + 2k_2 + 2k_3 + k_4)/6 = 4.006818970044454$$

Step 4 $t_4 = 2$

$$k_1 = hf(t_3, w_3) = 0.5f(1.5, 4.006818970044454) = 1.378409485022227$$

$$k_2 = hf(t_3 + h/2, w_3 + k_1/2) = 0.5f(1.75, 4.696023712555567) = 1.316761856277783$$

$$K_3 = hf(t_3 + h/2, w_3 + k_2/2) = 0.5f(1.75, 4.665199898183346) = 1.301349949091673$$

$$K_4 = hf(t_3 + h, w_3 + K_3) = 0.5f(2, 5.308168919136127) = 1.154084459568063$$

$$w_4 = w_3 + (k_1 + 2k_2 + 2k_3 + k_4)/6 = 5.301605229265987$$

Now let's compare what we got with the exact solution

t_i	Exact solution $y(t_i)$	Numerical solution w_i	Error $w_i - y(t_i)$
0.0	0.5	0.5	0
0.5	1.425639364649936	1.425130208333333	0.000509156316603
1.0	2.640859085770477	2.63960266113281	0.001256424637665
1.5	4.009155464830968	4.00681897004445	0.002336494786515
2.0	5.305471950534675	5.30160522926598	0.003866721268688

UNIT-5 (ADVANCE ENGINEERING MATHEMATICS)

MILNE'S PREDICTOR CORRECTOR METHODS

LECTURE NO: -45

Milne's Predictor Corrector Methods

Milne's simpson predictor formula is

$$y_{n+1,p} = y_{n-3} + 4h/3 (2y'_{n-2} - y'_{n-1} + 2y'_n)$$

putting $n=3$, we get

$$y_{4,p} = y_0 + 4h/3 (2y'_1 - y'_2 + 2y'_3)$$

Milne's simpson corrector formula is

$$y_{n+1,c} = y_{n-1} + h/3 (y'_{n-1} + 4y'_n + y'_{n+1})$$

putting $n=3$, we get

$$y_{4,c} = y_2 + h/3 (y'_2 + 4y'_3 + y'_4)$$

Examples

1. $y' = x + y^2$,

x_i	0	0.5	1	1.5
y_i	2	2.636	3.595	4.968

Find $y(2)$ by Milne's simpson predictor corrector method

Solution:

$$y' = x + y^2$$

Milne's simpson predictor formula is

$$y_{n+1,p} = y_{n-3} + 4h/3 (2y'_{n-2} - y'_{n-1} + 2y'_n)$$

putting $n=3$, we get

$$y_{4,p} = y_0 + 4h/3 (2y'_1 - y'_2 + 2y'_3) \rightarrow (2)$$

We have given that

$$x_0=0, x_1=0.5, x_2=1, x_3=1.5$$

and using runge kutta 4 method, we get

$$y_0=2, y_1=2.636, y_2=3.595, y_3=4.968$$

$$y' = x + y^2$$

$$y'_1 = x + y^2 = 1.568 \text{ (where } x=0.5, y=2.636)$$

$$y'_2 = x + y^2 = 2.2975 \text{ (where } x=1, y=3.595)$$

$$y'_3 = x + y^2 = 3.234 \text{ (where } x=1.5, y=4.968)$$

putting the values in (2), we get
 $y_4, p = y_0 + 4h^3(2y'_1 - y'_2 + 2y'_3) \rightarrow (2)$

$$y_4, p = 2 + 4 \cdot 0.53 \cdot (2 \cdot 1.568 - 2.2975 + 2 \cdot 3.234)$$

$$y_4, p = 6.871$$

So, the predicted value is 6.871

Now, we will correct it by corrector method to get the final value
 $y'_4 = x + y^2 = 4.4355 \text{ (where } x=2, y=6.871)$

Milne's simpson corrector formula is
 $y_{n+1, c} = y_{n-1} + h^3(y'_{n-1} + 4y'_n + y'_{n+1})$

putting $n=3$, we get

$$y_4, c = y_2 + h^3(y'_2 + 4y'_3 + y'_4)$$

$$y_4, c = 3.595 + 0.53 \cdot (2.2975 + 4 \cdot 3.234 + 4.4355)$$

$$y_4 = 6.8732$$

$$\therefore y(2) = 6.8732 \text{ (Answer)}$$

2. $y' = y - x^3, y(0) = 1$. Find $y(0.4)$ by Milne's simpson predictor corrector method

Solution:

$$y' = y - x^3$$

Milne's simpson predictor formula is
 $y_{n+1, p} = y_{n-3} + 4h^3(2y'_{n-2} - y'_{n-1} + 2y'_n)$

putting $n=3$, we get

$$y_4, p = y_0 + 4h^3(2y'_1 - y'_2 + 2y'_3) \rightarrow (2)$$

We have given that

$$x_0=0, x_1=0.1, x_2=0.2, x_3=0.3$$

and using runge kutta 4 method, we get

$$y_0=1, y_1=1.1051, y_2=1.221, y_3=1.3477$$

$$y' = y - x^3$$

$$y'_1 = y - x^3 = 1.1042 \text{ (where } x=0.1, y=1.1051\text{)}$$

$$y'_2 = y - x^3 = 1.213 \text{ (where } x=0.2, y=1.221\text{)}$$

$$y'_3 = y - x^3 = 1.3207 \text{ (where } x=0.3, y=1.3477\text{)}$$

putting the values in (2), we get

$$y_4, p = y_0 + 4h^3(2y'_1 - y'_2 + 2y'_3) \rightarrow (2)$$

$$y_4, p = 1 + 4 \cdot 0.1^3 \cdot (2 \cdot 1.1042 - 1.213 + 2 \cdot 1.3207)$$

$$y_4, p = 1.4849$$

So, the predicted value is 1.4849

Now, we will correct it by corrector method to get the final value

$$y'_4 = y - x^3 = 1.4209 \text{ (where } x=0.4, y=1.4849\text{)}$$

Milne's simpson corrector formula is

$$y_{n+1, c} = y_{n-1} + h^3(y'_{n-1} + 4y'_n + y'_{n+1})$$

putting $n=3$, we get

$$y_4, c = y_2 + h^3(y'_2 + 4y'_3 + y'_4)$$

$$y_4, c = 1.221 + 0.1^3 \cdot (1.213 + 4 \cdot 1.3207 + 1.4209)$$

$$y_4 = 1.4849$$

$$\therefore y(0.4) = 1.4849 \text{ (Answer)}$$